PHYS 7210 — Graduate Quantum Mechanics

Wave Functions & Measurement

Lecture 12 — Course Instructor

Department of Physics · Spring 2026

Today's Topics

01 Wave-Particle Duality

02 The Schrödinger Equation

03 Probability Densities

04 The Uncertainty Principle

05 Quantum Tunneling

Wave-Particle Duality

Louis de Broglie proposed that all matter exhibits both wave-like and particle-like properties. The wavelength associated with a particle depends on its momentum.

Wave → Particle
(click to morph)



$$E = h\nu$$

Energy-frequency

$$\lambda = \frac{h}{p}$$

de Broglie

The Schrödinger Equation

$$\hat{H}\,\psi = E\,\psi \qquad \text{(Time-independent Schrödinger equation)}$$

Click → to expand the derivation

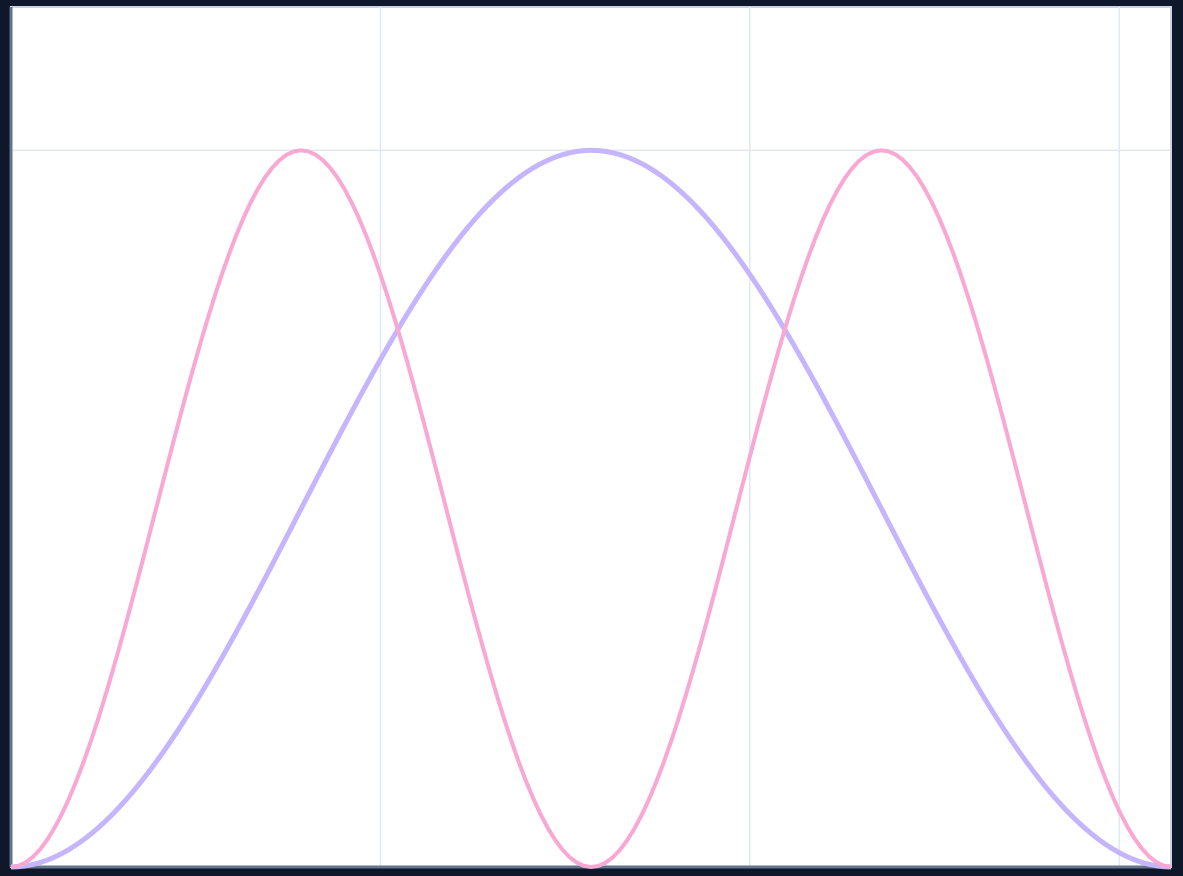
The Hamiltonian operator $\hat{H}$ encodes the total energy of the system — kinetic plus potential.

$\Psi(\vec{r}, t)$ — wave function
encodes all quantum state information

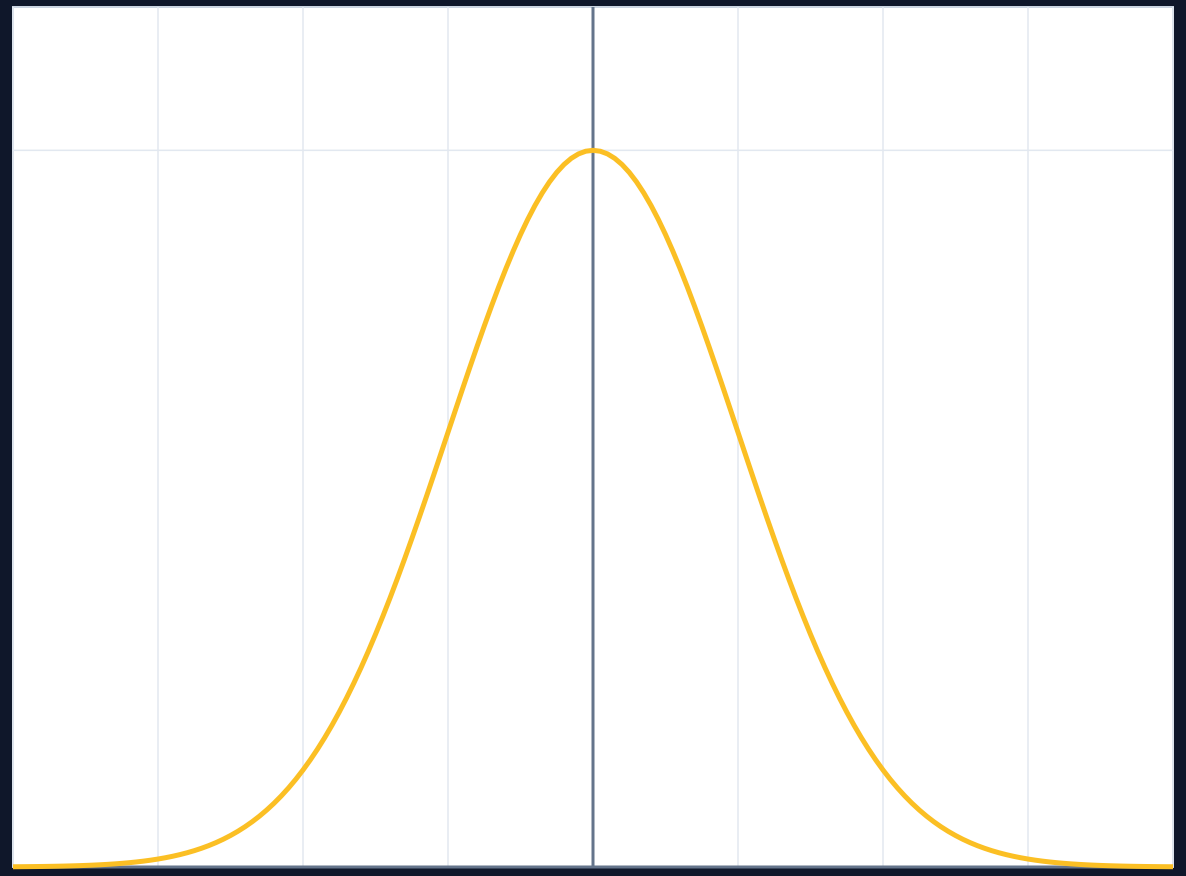
$\hbar = h/2\pi$ — reduced Planck constant
$\hbar \approx 1.055 \times 10^{-34}$ J·s

$V(\vec{r})$ — potential energy field
determines bound state solutions

Probability Densities



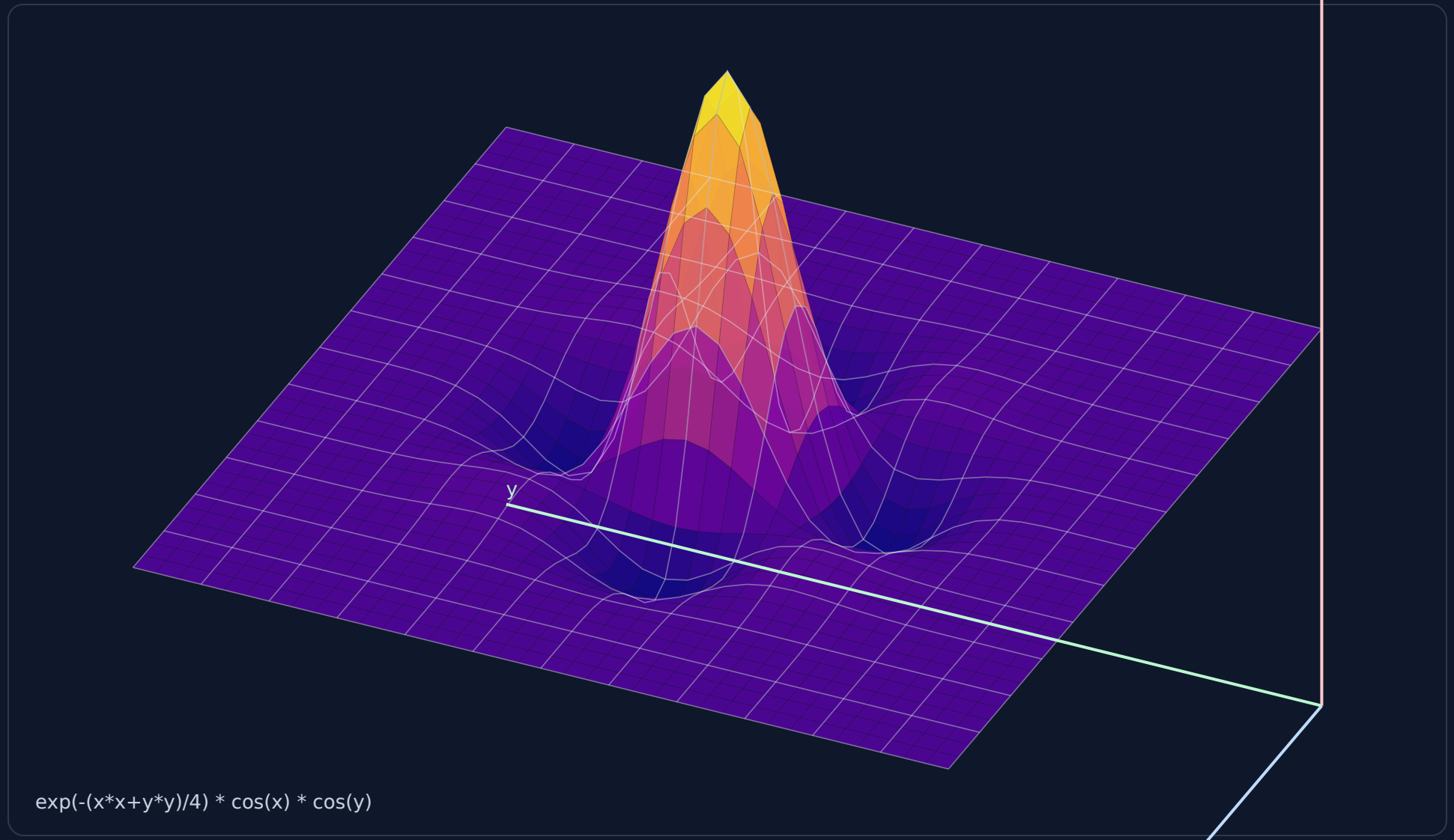
Particle in a box — quantum number n = 1, 2



Gaussian wave packet — minimum uncertainty state

$$P(a,b) = \int_a^b |\Psi(x,t)|^2 \, dx$$

Atomic Orbital Probability



Probability amplitude |ψ(′r)|² for a hydrogen-like orbital — surface height encodes probability density

The Uncertainty Principle

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

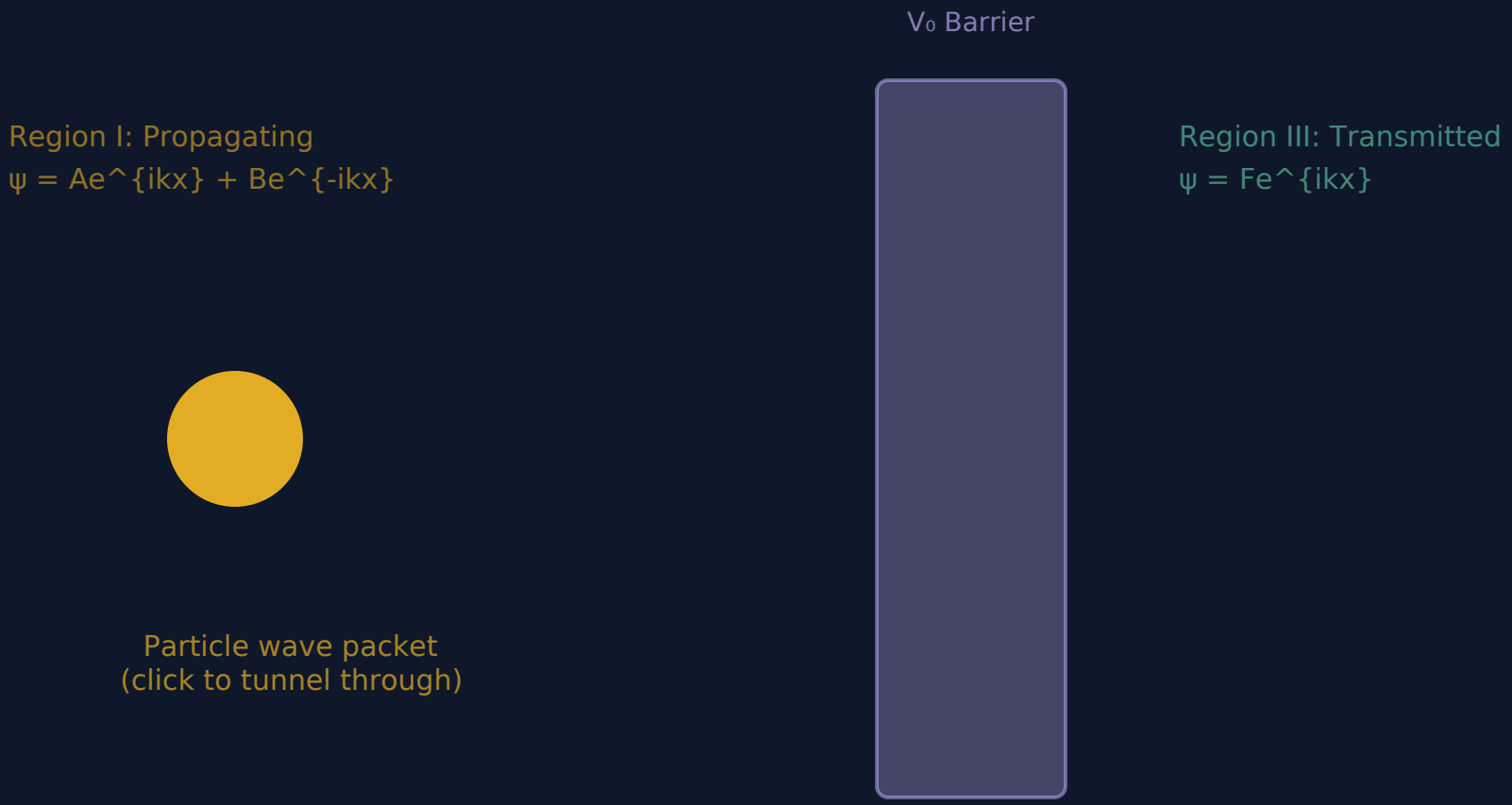
We cannot simultaneously know both the exact position and momentum of a particle — this is not a limitation of o

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

$$\sigma_A \sigma_B \geq \frac{1}{2} | \langle [\hat{A}, \hat{B}] \rangle |$$

Robertson generalization — any two observables

Quantum Tunneling



$$T \approx e^{-\kappa L}, \kappa = \sqrt{\frac{2m(V - E)}{\hbar}}$$

Transmission probability decays exponentially with barrier width L and the evanescent decay constant κ

Key Takeaways

01

Wave functions encode all measurable information about a quantum system

02

Schrödinger equation governs the time evolution of quantum states

03

Measurement collapses the wave function — Born rule gives the probability

04

Uncertainty is fundamental to nature — not a limitation of instruments

Questions?

PHYS 7210 — Office hours: Tue/Thu 2–4 pm, Room 412

Next lecture: Quantum Entanglement & Bell's Theorem

Instructor · Department of Physics · Spring 2026